

Scale and Cost Estimation for the HeartCompass System

1. Audit Background and Estimation Methodology

1.1 Target System and Boundary Definition

The audit target of this report is the **HeartCompass (Digital Immortality System)**, a core repository project developed during the course, rather than the basic CRM example provided in the lab guidelines. To ensure a precise and objective valuation, the system boundary is strictly defined as follows:

- **Internal Application Boundary (Counted Assets):** Includes the CLI administration terminal, the Lark Bot asynchronous gateway, the dual-track Agent graph engines (FRBuildingGraph / ConversationGraph), and the underlying relational persistence database.
- **External External Entities (Exclusions):** Includes the Volcano Ark LLM API services and the Lark Open Platform message event triggers.

1.2 Dual-Track Cross-Validation Model

Given that HeartCompass is an innovative AI prototype heavily leveraging existing ecological frameworks rather than a standard commercial software built from scratch, relying on a single traditional estimation model would lead to heavily distorted financial metrics. Therefore, this report implements a **Dual-Track Cross-Validation Framework**:

1. **Primary Metric (WBS-Driven Backward Tracking Method):** Utilizing the actual engineering logs from the repository (Biweekly Report #4 records a total cumulative input of 220 hours with a weighted completion rate of 94.8%), this serves as the definitive financial cost baseline.
2. **Counterparty Metric (NESMA Logic Transaction Measurement):** Conducting a rigorous inventory of the system's data functions (ILF/ELF) and transactional functions (EI/EO/EQ) purely from an end-user-recognizable perspective. This serves as an analytical check for R&D efficiency rather than a commercial quote.

1.3 Baseline Pricing Parameters and Core Formulas

The labor cost calculations are strictly aligned with the regional baseline for Beijing in the *2024 China Software Industry Benchmark Data*:

- **Regional Software Development Labor Rate:** 32,129 RMB/Person-Month

- **Statutory Monthly Working Hours:** 22.5 Days/Month × 8 Hours/Day = 180 Person-Hours/Person-Month
- **Blended Hourly Labor Rate:** 32,129 ÷ 180 = 178.49 RMB/Hour

2. Architectural Asset Mapping and Data Flow Inventory

Prior to granular counting, a static engineering audit of the source repository was conducted to map the core service boundaries and data entity assets:

Asset Code	Service Boundary	Repository Source & Document Reference	Core Interaction & Data Flow
SRV-01	CLI Administration Terminal	auth.py , fr.py , lark_service.py	Handles environment initialization, system diagnostic checks, user authentication, and local persona synchronization scheduling.
SRV-02	Lark Bot Async Gateway	index.py , menu.py	Intercepts Lark event streams, routes instant menu commands, switches active personas, and pushes rich asynchronous cards.
AGT-01	Asynchronous Persona Builder	BRIEF_INTRO.md	Consumes raw source text materials, executes fine-grained entity extraction, conflicts checking, and compiles persona completeness reports.
AGT-02	Multi-Turn Conversation Engine	BRIEF_INTRO.md	Loads active personas, triggers semantic memory recall, maintains rolling short-term context summaries, and renders multi-turn chat messages.
DB-01	Persistence Data Layer	src/database/ models.py	Materializes core schemas including User, FigureAndRelation, OriginalSource, Knowledge, and operational graph logs.
WBS-01	R&D Baseline Tracking	biweekly_report_4_final.md	Captures the historical R&D footprints of 4 team members across 12 distinct functional development sprints.

3. Software Scale Measurement via the NESMA Framework

This section strictly follows the NESMA Function Point Analysis (FPA) standard to quantify the functional size within the system boundary. Purely internal technical operations have collapsed to prevent artificial scale inflation:

3.1 Data Functions (Information Asset Inventory)

- **ILF - User Authentication Registry (Low Complexity, 7 FP):** Maintains local User accounts, active Auth states, and internal mappings to external Lark `open_id` credentials.
- **ILF - Profile & Long-Term Memory Vault (Medium Complexity, 10 FP):** Persists FigureAndRelation records, granular Feeds, Knowledge entities, and conflict/system update logs.
- **ILF - Short-Term Session Context Cache (Low Complexity, 7 FP):** Coordinates the rolling session summaries, conversation messages, and state checkpoints within the ConversationGraph.
- **ELF - Lark Platform External Interfaces (Low Complexity, 5 FP):** Tracks user profiles, structural message payloads, and interactive card components hosted and managed by the external Lark ecosystem.

3.2 Transactional Functions (Triggered Business Processes)

- **EI - Environment Bootstrapping & Provisioning (Medium Complexity, 4 FP):** Executes CLI commands including system setup, doctor checks, user registrations, and account-to-platform linkage.
- **EI - Asynchronous Persona Compiling Trigger (High Complexity, 6 FP):** Captured via the `/build_persona` directive which spins up the complete FRBuildingGraph workflow and populates downstream databases.
- **EO - Multi-Turn Algorithmic Chat Responses (High Complexity, 7 FP):** Orchestrates the dynamic output derived from the ConversationGraph execution loop to compile the final `messages_to_send` payload.
- **EO - Persona Rendered Visual Cards (Medium Complexity, 5 FP):** Transforms raw persona updates, profile metrics, and markdown blocks into rich user-facing Lark interactive messages.
- **EQ - Instant Navigation Menu & State Interrogation (Low Complexity, 3 FP):** Processes real-time platform shortcodes such as `/menu`, `/list_available_persons`, and `/whoami` queries.

3.3 Functional Size Metrics Summary

HeartCompass MVP Functional Scale = $7 + 10 + 7 + 5 + 4 + 6 + 7 + 5 + 3 = 54 \text{ FP}$

Scale Recalibration Note: In this refined measurement, fine-grained technical assets (such as separate conflict tracking logs, intermediate execution graphs, and local service calls) from earlier draft iterations have collapsed into cohesive, end-user-perceivable transactions. This confines the MVP scope to a highly realistic and grounded **54 FP**.

4. Financial Valuation and Multidimensional Validation

4.1 Primary Valuation: WBS-Driven Backward Tracking Calculation

Extrapolating from the actual project engineering baselines captured in the repository:

- **Actual Hours Incurred to Date:** 220 Person-Hours
- **Weighted R&D Progress Baseline:** 94.8%
- **Projected Final Project Effort:** $\text{Total Effort} = \frac{220}{0.948} = 232.07 \text{ Person-Hours}$
- **Total Estimated System Development Cost:**
 $\text{Total Cost} = 232.07 \text{ Hours} \times 178.49 \text{ RMB/Hour} = 41422.76 \text{ RMB (approx. 41.4K RMB)}$
- **Theoretical Project Schedule Forecast:** Assuming full parallel operations within a 4-person engineering team (8 hours/day, 22.5 days/month), the theoretical calendar schedule for the calculated effort is: $232.07 \div 4 \div 8 \div 22.5 = 0.32 \text{ Months}$

4.2 Independent Cross-Check: Realized Hours Direct Settlement

To isolate the calculation from cascading estimation errors, a direct cash-settlement analysis based on actual realized hours was performed:

- **Direct Cost of Currently Completed R&D Milestones:**
 $220 \text{ Hours} \times 178.49 \text{ RMB/Hour} = 39268.78 \text{ RMB}$
- **Discrete Variance Analysis Between Both Modalities:**
 $\text{Variance Rate} = \frac{|41422.76 - 39268.78|}{41422.76} \times 100\% = 5.20\%$

Model Reliability Validation: The resulting financial discrepancy between the two time-tracking methodologies is locked at a minor **5.20%**. This falls well within the standard $\pm 20\%$ engineering error envelope, validating the high internal consistency of the WBS-driven valuation model.

4.3 Deviation Analysis Against Commercial Baseline

For a broader perspective, applying the cross-industry average R&D productivity standard (6.83 Hours/FP) directly to the system's 54 FP would yield a commercial outsourcing quote of roughly **65.9K RMB**.

This report explicitly rejects using that commercial valuation as the submission baseline. The reason is that HeartCompass's R&D cycle heavily relied on highly mature ecosystems, including LangGraph orchestration, Volcano Ark LLM platforms, the Lark SDK, and PostgreSQL/pgvector plugins. This high "ecological reuse factor" allowed the team to completely bypass rebuilding basic architectural building blocks. Consequently, the actual WBS hours are significantly lower than cross-industry benchmarks, making the WBS-driven tracking method the only accurate reflection of the academic prototype's true cost.

5. Software Engineering Economics Insights

Re-evaluating the scale metrics and performing reverse financial modeling on HeartCompass yielded several valuable insights into software engineering economics:

1. **The Essence of the User-Perceived Boundary:** During the initial phases of the estimation, it is incredibly easy to mistake low-level technical artifacts (like individual Prompt configurations, standalone graph nodes, or intermediate schema helpers) for separate function points, which artificially blows up the system size. The NESMA framework forces engineers to strip away internal code implementation details and evaluate the software entirely from an end-user perspective. This provides a vital check against scope creep.
2. **The Economic Impact of Ecological Reuse:** HeartCompass is a highly advanced, multi-turn agent system driven by a CLI and Lark Bot interface. However, because the development was built on top of robust open-source frameworks and cloud APIs, the actual labor hours (220h) are much lower than commercial industry averages. This proves that in modern software engineering, smart architectural selection and framework reuse are the most powerful levers for optimizing cost and efficiency.
3. **WBS Logs as the Core Anchor for Cost Control:** While static function point analysis provides an excellent theoretical baseline, dynamic WBS logs represent the ultimate source of truth for software asset valuation. By setting up dual-track checks and examining the variance, engineers can move beyond mere guessing and use objective data to justify cost structures to stakeholders.

6. Audit References and Source Documentation

1. *Case_Software Size Estimation (NESMA).pdf* — Comprehensive Functional Size Standards.
2. *Experiment Instruction+Exp 2-2_Software Cost Estimation.pdf* — Lab Guidelines for Software Economics.
3. <https://github.com/Jackey0903/HeartCompass> — HeartCompass Core Codebase and Source Assets.
4. `HeartCompass/README.md` — Core System Operations and Bootstrapping Manual.
5. `HeartCompass/docs/BRIEF_INTRO.md` — LLM Graph Topologies, Agent Architectures, and Schema Layouts.
6. `HeartCompass/src/database/models.py` — Database Schema Definitions and Object Relational Mapping.
7. `HeartCompass/src/channels/lark/integration/` — Lark Communication Routers and Message Mapping Handlers.
8. `HeartCompass/docs/biweekly_report_4_final.md` — WBS Real-Time Effort Metrics and Team Workload Baselines.

9. *2024 China Software Industry Benchmark Data (CSBMK®-202410)* — Standard National Labor Rates.